

Boss Challenge - 100 Toughest MCQs (Class 7)

Acids, Bases & Salts | Electric Current & its Effects | Arithmetic Expressions (order of operations)

Each question has exactly ONE correct option. Marking: +4 correct, -1 wrong, 0 blank. Write A/B/C/D on a separate answer sheet. (Answer key provided separately.)

1. A liquid turns blue litmus red and has no effect on red litmus. What can you be sure about it?
(A) It is a base
(B) It is an acid
(C) It is neutral
(D) It is salt water
2. In a circuit, one bulb glows. You add a second identical cell in the same direction. What happens to the bulb?
(A) It glows dimmer
(B) It glows brighter
(C) It stops glowing
(D) No change at all
3. Find the value: $12 - 3 \times (4 - 6)$
(A) 18
(B) 6
(C) -18
(D) 30
4. Turmeric paste (yellow) is rubbed on a cloth. A drop of soap solution falls on it and the spot turns red. This tells us soap solution is:
(A) acidic
(B) basic
(C) neutral
(D) a salt
5. Two bulbs are in a single loop (series) with one cell. One bulb's filament breaks. The other bulb:
(A) glows brighter
(B) glows the same
(C) goes off
(D) glows dimmer
6. Which bracket placement makes this true: $8 - 2 + 3 = 3$?
(A) $8 - (2 + 3)$
(B) $(8 - 2) + 3$
(C) $8 - (2 - 3)$
(D) $(8 - 2 + 3)$

7. A farmer's soil is too acidic. Which should he add?
- (A) Vinegar
 - (B) Lemon juice
 - (C) Quick lime (a base)
 - (D) More acid
8. The heating element of a room heater is made of a special wire instead of copper because the wire must:
- (A) be cheap
 - (B) get very hot without melting
 - (C) bend easily
 - (D) stay cool
9. Without fully solving, which is larger? $P = 25 \times 8$ and $Q = 25 \times (5 + 3)$
- (A) P is larger
 - (B) Q is larger
 - (C) They are equal
 - (D) Cannot say
10. An ant's sting leaves an acid on the skin. The best home remedy is to rub:
- (A) vinegar
 - (B) lemon juice
 - (C) baking soda solution (a mild base)
 - (D) curd
11. The wire inside an electric fuse melts when:
- (A) too little current flows
 - (B) too much current flows
 - (C) the switch is off
 - (D) the bulb is new
12. Evaluate: $-5 + (-3) \times 2$
- (A) -16
 - (B) -11
 - (C) 4
 - (D) -1
13. Equal amounts of an acid and a base are mixed in correct proportion. The final mixture will most likely turn litmus:
- (A) red
 - (B) blue
 - (C) neither (no change)
 - (D) first red then blue

14. When current passes through a coil of wire wrapped on an iron nail, the nail can pick up pins. Switch off the current and the pins fall. This shows the nail became:

- (A) a permanent magnet
- (B) a temporary (electro)magnet
- (C) an acid
- (D) a fuse

15. Which expression has the same value as $6 \times 9 + 6 \times 1$?

- (A) 6×10
- (B) 6×91
- (C) 12×10
- (D) $6 + 9 + 1$

16. Distilled water does not change the colour of litmus. This is because distilled water is:

- (A) a strong acid
- (B) a strong base
- (C) neutral
- (D) a salt solution

17. To make an electromagnet stronger, you should:

- (A) use fewer turns of wire
- (B) use a wooden rod instead of iron
- (C) increase the number of turns of wire
- (D) reduce the current

18. The number line jump: start at 3, the expression is $3 - 7 + 2$. Where do you land?

- (A) -2
- (B) 8
- (C) -6
- (D) 2

19. Lemon juice is sour. Which substance, when added in the right amount, will remove its sourness?

- (A) vinegar
- (B) baking soda (a base)
- (C) tamarind
- (D) common salt

20. In a simple circuit the bulb does NOT glow even though the cell is new and wires are joined. The most likely reason is:

- (A) the switch is open (off)
- (B) there are too many wires
- (C) the cell is too strong
- (D) the bulb is too small

21. Which of these is the correct reading of "subtract 4 from the sum of 9 and 6"?

- (A) $4 - (9 + 6)$
- (B) $(9 + 6) - 4$
- (C) $9 + (6 - 4)$
- (D) $9 - 6 + 4$

22. China rose solution turns dark pink with acids and green with bases. A liquid turns it green. The liquid is:

- (A) acidic
- (B) basic
- (C) neutral
- (D) distilled water

23. An electric bell rings continuously because the electromagnet in it switches:

- (A) on once only
- (B) off once only
- (C) on and off again and again
- (D) never

24. Find the missing number: $4 \times (\blacksquare - 2) = 20$

- (A) 3
- (B) 5
- (C) 7
- (D) 22

25. Milk of magnesia is taken for an acidic stomach. This works by:

- (A) adding more acid
- (B) neutralising the extra acid
- (C) cooling the stomach
- (D) drying the stomach

26. Two circuits use the same bulb and wires. Circuit X has 1 cell; Circuit Y has 3 cells in series (same direction). Which is correct?

- (A) X is brighter
- (B) Y is brighter
- (C) both equal
- (D) Y does not glow

27. Which one is FALSE about the expression $10 - 2 \times 3$?

- (A) Multiplication is done first
- (B) Its value is 4
- (C) Its value is 24
- (D) The answer is positive

28. Acids taste sour and bases taste bitter. We should NEVER taste lab chemicals to test them. Instead we use:

- (A) our tongue carefully
- (B) indicators
- (C) a magnet
- (D) a thermometer

29. A compass needle near a current-carrying wire moves. When the current is switched off, the needle:

- (A) keeps moving
- (B) returns to its normal direction
- (C) melts
- (D) becomes an acid

30. Which gives the SAME answer as $100 - (40 + 25)$?

- (A) $100 - 40 + 25$
- (B) $100 - 40 - 25$
- (C) $100 + 40 - 25$
- (D) $40 + 25 - 100$

31. A gardener finds his soil too basic (alkaline). Which is the best material to add?

- (A) Lime
- (B) Baking soda
- (C) Decaying organic matter / compost (slightly acidic)
- (D) Washing soda

32. In which device is the heating effect of current the MAIN purpose?

- (A) Electric bell
- (B) Electric iron
- (C) Magnetic crane
- (D) Compass

33. True or false sorting - which statement is TRUE? "Adding the same brackets that don't change grouping does not change the value."

- (A) $7 + (3 + 2) = (7 + 3) + 2$ is true
- (B) $7 - (3 - 2) = (7 - 3) - 2$ is true
- (C) both are true
- (D) both are false

34. Phenolphthalein is colourless in acids and pink in bases. You add it to a liquid and it stays colourless. The liquid is:

- (A) definitely a base
- (B) acidic or neutral
- (C) only neutral
- (D) only salt

35. A fuse of low rating is replaced by a thick copper wire by mistake. The danger is:
- (A) the circuit will never work
 - (B) it will not melt during overload, risking fire
 - (C) the bulb will glow blue
 - (D) nothing, it is safer
36. Evaluate: $(-2) \times (-3) \times (-1)$
- (A) 6
 - (B) -6
 - (C) 5
 - (D) -5
37. Vinegar is added drop by drop to a base with red litmus already turned blue. After enough vinegar, the litmus may turn:
- (A) stay blue forever
 - (B) red, once the base is neutralised and acid is in excess
 - (C) green
 - (D) yellow
38. Why is the coil of an electromagnet usually wound on soft iron and not steel?
- (A) Steel is cheaper
 - (B) Soft iron loses its magnetism quickly when current stops
 - (C) Steel cannot be magnetised
 - (D) Soft iron is a good acid
39. Insert one pair of brackets to make it true: $2 + 3 \times 4 = 20$
- (A) $(2 + 3) \times 4$
 - (B) $2 + (3 \times 4)$
 - (C) $(2 + 3 \times 4)$
 - (D) impossible
40. Which natural material can act as an indicator?
- (A) Common salt
 - (B) Red cabbage / China rose / turmeric
 - (C) Sugar
 - (D) Sand
41. A magnetic crane in a scrap yard uses an electromagnet rather than a normal magnet because it can:
- (A) never be switched off
 - (B) be switched on to lift and off to drop the load
 - (C) lift only plastic
 - (D) work without electricity

42. Compare without solving: $A = 15 + 15 + 15$ and $B = 15 \times 3$. Then:

- (A) $A > B$
- (B) $A < B$
- (C) $A = B$
- (D) cannot say

43. When an acid reacts with a base, the products are always:

- (A) acid and water
- (B) salt and water
- (C) two acids
- (D) a gas only

44. In a torch, the cells must be placed:

- (A) any way you like
- (B) with the same terminals touching (+ to +)
- (C) so that + of one touches - of the next
- (D) without touching each other

45. The value of $36 \div (3 + 3) \times 2$ is:

- (A) 2
- (B) 12
- (C) 6
- (D) 3

46. Soap is slippery and turns red litmus blue. So soap is:

- (A) acidic
- (B) basic
- (C) neutral
- (D) a strong acid

47. Three bulbs glow in one loop with a battery. You unscrew (remove) the middle bulb. The other two bulbs:

- (A) glow brighter
- (B) glow the same
- (C) go off
- (D) glow once then off

48. Which statement about the expression $5 \times (2 + 3)$ is correct?

- (A) It equals $5 \times 2 + 3 = 13$
- (B) It equals $5 \times 2 + 5 \times 3 = 25$
- (C) It equals $5 + 2 + 3 = 10$
- (D) Brackets can be ignored

49. Tooth decay starts when bacteria in the mouth make acid. Using toothpaste (usually basic) helps because it:

- (A) adds acid
- (B) neutralises the acid
- (C) cools the teeth
- (D) makes more bacteria

50. The amount of heat produced in a wire by current depends on the wire's material and on:

- (A) its colour
- (B) the current passing through it
- (C) the time of day
- (D) the room's smell

51. Fill the blank to make it true: $48 \div (\blacksquare - 4) = 8$

- (A) 2
- (B) 6
- (C) 10
- (D) 12

52. A solution turns blue litmus red AND turns phenolphthalein colourless. The solution is:

- (A) basic
- (B) acidic
- (C) neutral
- (D) salt and water only

53. Why are electric wires inside homes covered with plastic or rubber?

- (A) to look colourful
- (B) because plastic/rubber does not let current pass (insulator), keeping us safe
- (C) to make the wire heavy
- (D) to make it an acid

54. The expression "double the number, then add 5" for a number n is:

- (A) $2 + n + 5$
- (B) $2n + 5$
- (C) $2(n + 5)$
- (D) $n + 5 + n + 5$

55. Curd and lemon are both sour because they contain:

- (A) bases
- (B) acids
- (C) salts only
- (D) neutral water

56. A modern house uses an MCB instead of a fuse mainly because an MCB:
- (A) is an acid
 - (B) can be switched back on after the fault is fixed (no wire to replace)
 - (C) gives more current
 - (D) never breaks the circuit
57. Which two expressions are equal? I) $20 - (5 + 4)$ II) $20 - 5 - 4$ III) $20 - 5 + 4$
- (A) I and II
 - (B) I and III
 - (C) II and III
 - (D) all three
58. When baking soda (a base) reacts with vinegar (an acid), bubbles form. The mixture, after reaction, is mostly:
- (A) a strong acid
 - (B) a strong base
 - (C) close to neutral with a salt and gas
 - (D) pure metal
59. A wire carrying current is brought near a small magnetic compass. The clearest proof that current has a magnetic effect is:
- (A) the wire melts
 - (B) the compass needle deflects (turns)
 - (C) the bulb glows
 - (D) the wire becomes sour
60. Evaluate: $3 \times 4 + 3 \times 6 - 3 \times 2$
- (A) 24
 - (B) 30
 - (C) 18
 - (D) 3×8
61. Rainwater is normally slightly acidic. Near factories it can become strongly acidic. This polluted rain is called:
- (A) heavy rain
 - (B) acid rain
 - (C) salt rain
 - (D) base rain
62. In an electromagnet, two ways to make it lift MORE pins are:
- (A) fewer turns and less current
 - (B) more turns and more current
 - (C) a plastic core and less current
 - (D) more turns but no current

63. Which is the correct value of $7 + 7 \div 7 + 7 \times 7$?

- (A) 57
- (B) 8
- (C) 105
- (D) 14

64. A liquid does NOT change blue litmus, does NOT change red litmus, and tastes neither sour nor bitter. It is most likely:

- (A) an acid
- (B) a base
- (C) water (neutral)
- (D) a strong indicator

65. The element (coil) of an electric heater glows red-hot but its connecting copper wires stay cool. This is because the coil:

- (A) is thinner and made of a special metal that resists current more, producing heat
- (B) is made of plastic
- (C) carries no current
- (D) is an acid

66. Which placement of brackets makes the value the LARGEST? Using $16 - 4 - 4$

- (A) $(16 - 4) - 4$
- (B) $16 - (4 - 4)$
- (C) both equal
- (D) neither has brackets

67. "Olfactory indicators" like onion or vanilla are used to test acids and bases by their:

- (A) colour
- (B) taste
- (C) smell (which changes)
- (D) weight

68. If a bulb and a buzzer are in the same single loop and the bulb's filament breaks, the buzzer will:

- (A) get louder
- (B) stay the same
- (C) stop
- (D) ring twice

69. Evaluate: $(8 + 4) \div 2 - 3 \times (1 + 1)$

- (A) 0
- (B) 6
- (C) 12
- (D) -3

70. A factory's waste water is acidic. Before releasing it into a river, the factory should treat it with a base to:

- (A) make it more acidic
- (B) neutralise it
- (C) heat it
- (D) colour it

71. Increasing the number of cells (in the same direction) in a circuit increases the:

- (A) length of the wire
- (B) current, making bulbs brighter
- (C) colour of the bulb
- (D) smell

72. Which sentence describes $24 \div 6 + 2$ correctly?

- (A) Divide first, then add: answer 6
- (B) Add first, then divide: answer 3
- (C) Answer is 24
- (D) Answer is $8 \div 2$

73. A gas turns moist red litmus blue. The gas is most likely:

- (A) acidic
- (B) basic
- (C) neutral
- (D) a salt

74. The main reason a thin fuse wire is used (and not a thick one) is that a thin wire:

- (A) melts at a safe lower current to break the circuit
- (B) carries more current
- (C) never melts
- (D) is a base

75. Without solving fully, which is greater? $A = 99 \times 99$ and $B = 99 \times 100 - 99$

- (A) $A > B$
- (B) $A < B$
- (C) $A = B$
- (D) cannot say

76. Both an acid and a base, when strong, can harm the skin. The correct first step if acid falls on your hand is to:

- (A) rub it harder
- (B) wash with plenty of water at once
- (C) add a strong base directly
- (D) leave it

77. Which device would NOT work using only the magnetic effect of current?
- (A) Electric bell
 - (B) Electromagnet crane
 - (C) Electric room heater (its main job)
 - (D) A coil deflecting a compass
78. Evaluate: $-10 - (-4) + (-6)$
- (A) -12
 - (B) -20
 - (C) 0
 - (D) -8
79. Phenolphthalein is pink in a liquid. You slowly add lemon juice. The pink colour:
- (A) becomes darker pink
 - (B) fades and disappears as it becomes acidic/neutral
 - (C) turns blue
 - (D) does not change
80. A switch in a circuit is a device that:
- (A) makes current sour
 - (B) opens or closes the circuit to stop or allow current
 - (C) heats the room
 - (D) acts as an acid
81. Two friends evaluate $18 - 6 \div 3$. Anu says 4, Ravi says 16. Who is right and why?
- (A) Anu - subtract first
 - (B) Ravi - divide first ($6 \div 3 = 2$), then $18 - 2 = 16$
 - (C) Both are wrong
 - (D) Anu - answer is 4 by BODMAS
82. Baking soda is also used in fire extinguishers because, with an acid, it produces a gas that:
- (A) feeds the fire
 - (B) does not support burning and helps put out fire
 - (C) is an acid itself
 - (D) melts the fuse
83. If you reverse the direction of the cell (battery) in a simple circuit with a bulb, the bulb will:
- (A) not glow at all
 - (B) still glow (a bulb glows either way)
 - (C) explode
 - (D) become a magnet

84. Which expression equals 0 for every value of the number n ?

- (A) $n + n$
- (B) $n - n$
- (C) $n \times n$
- (D) $n + 0$

85. A drop of soap water is put on a turmeric-stained cloth (the stain went red). To turn the red turmeric stain back to yellow, you should add:

- (A) more soap
- (B) a base
- (C) an acid (like vinegar)
- (D) salt

86. The heating effect is unwanted (wasteful) in which appliance?

- (A) Electric heater
- (B) Electric iron
- (C) Wires carrying current to a fan (heat is a loss)
- (D) Toaster

87. Place brackets to make it true: $24 - 8 \div 4 + 2 = 6$

- (A) $24 - (8 \div 4) + 2$
- (B) $(24 - 8) \div 4 + 2$
- (C) $24 - 8 \div (4 + 2)$
- (D) $(24 - 8) \div (4 + 2)$

88. "Salt" in chemistry means:

- (A) only the common salt we eat
- (B) any product (besides water) formed when an acid reacts with a base
- (C) any sour substance
- (D) any bitter substance

89. The brightness of a bulb in a torch slowly reduces over days of use. The most likely reason is:

- (A) the bulb is becoming an acid
- (B) the cells are getting used up (weaker)
- (C) the switch is broken
- (D) the wire turned to iron

90. Evaluate: $2 \times [3 + (8 - 5)]$

- (A) 12
- (B) 11
- (C) 16
- (D) 6

91. Which household item is a base?

- (A) Lemon juice
- (B) Vinegar
- (C) Window-cleaning ammonia / soap solution
- (D) Curd

92. An electromagnet and a bar magnet both pick up nails. The KEY difference is that the electromagnet:

- (A) is heavier
- (B) can be turned on and off with a switch
- (C) is always weaker
- (D) needs no current

93. Which two are equal? I) $6 \times (10 - 1)$ II) $6 \times 10 - 6 \times 1$ III) $6 \times 10 - 1$

- (A) I and II
- (B) II and III
- (C) I and III
- (D) all three

94. When an acidic solution is added little by little to a basic solution, the temperature of the mixture often:

- (A) does not change at all
- (B) rises a little (neutralisation can release heat)
- (C) drops to freezing
- (D) becomes negative

95. A nichrome coil is used in heaters instead of copper mainly because nichrome:

- (A) is a magnet
- (B) resists current more and has a high melting point, so it heats up and glows safely
- (C) is an acid
- (D) carries no current

96. Find n if $3 \times n - 4 = 2 \times n + 5$

- (A) 1
- (B) 9
- (C) -9
- (D) 5

97. A wet finger touching a switch can be dangerous because water (with impurities):

- (A) is a base
- (B) can conduct electricity, giving a shock
- (C) stops all current
- (D) is an indicator

98. Which list is in order from acidic -> neutral -> basic?

- (A) soap, water, lemon juice
- (B) lemon juice, water, soap
- (C) water, lemon juice, soap
- (D) lemon juice, soap, water

99. Evaluate carefully: $50 - [20 - \{10 - (5 - 2)\}]$

- (A) 37
- (B) 23
- (C) 17
- (D) 43

100. A single circuit has 1 cell and 1 bulb (bulb glows). Three changes are made one at a time: (i) add a second cell in series, (ii) add a second bulb in the same loop, (iii) open the switch. The brightness order of the original bulb, from BRIGHTEST to OFF, is:

- (A) (i) brighter, (ii) dimmer, (iii) off
- (B) (ii) brighter, (i) dimmer, (iii) off
- (C) (iii) brighter, (i) off, (ii) dimmer
- (D) all three make it brighter